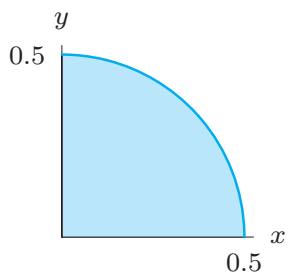
Suggested problems: 1-21 odd

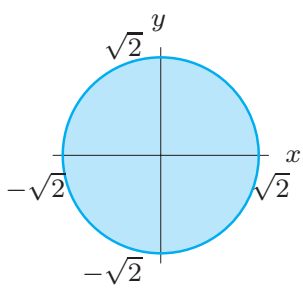
Problem 1: For each shaded region R below, write $\int_R f\ dA$ as an iterated integral in polar coordinates.

(a)



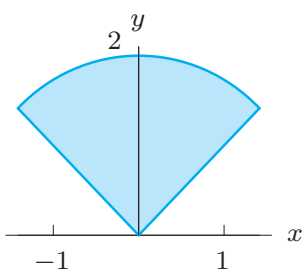
$\int_0^{\pi/2} \int_0^{1/2} f(r,\theta)\ r dr d\theta$

(b)



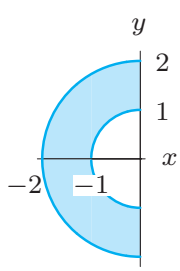
$\int_0^{2\pi} \int_0^{\sqrt{2}} f(r,\theta)\ r dr d\theta$

(c)



$\int_{\pi/4}^{3\pi/4} \int_0^{2} f(r,\theta)\ r dr d\theta$

(d)



$\int_{\pi/2}^{3\pi/2} \int_1^{2} f(r,\theta)\ r dr d\theta$

Problem 2: For each part below, sketch the region of integration.

(a) $\int_0^4 \int_{-\pi/2}^{\pi/2} f(r,\theta)\ r\ d\theta\ dr$ See class notes

(b) $\int_0^{\pi/4} \int_0^{1/\cos\theta} f(r,\theta)\ r\ dr\ d\theta$ See class notes

(c) $\int_{\pi/4}^{\pi/2} \int_0^{2/\sin\theta} f(r,\theta)\ r\ dr\ d\theta$ See class notes

Problem 3: Evaluate

$$\int_R \sqrt{x^2+y^2}\ dA$$

where R is $4 \le x^2+y^2 \le 9$.

$$\int_R \sqrt{x^2+y^2}\ dA = \int_2^3 \int_0^{2\pi} r^2\ d\theta dr = 2\pi \int_2^3 r^2\ dr = \frac{38\pi}{3}.$$

Problem 4: Evaluate

$$\int_R \sin\left(x^2+y^2\right)\ dA$$

where R is the disk of radius 2 centered at the origin.

$$\int_R \sin\left(x^2+y^2\right)\ dA = \int_0^2 \int_0^{2\pi} r\sin\left(r^2\right)\ d\theta dr = 2\pi \int_0^2 r\sin\left(r^2\right)\ dr = -\pi\cos\left(r^2\right)\Big|_0^2 = -\pi\cos(4)+\pi.$$

Problem 5: Convert the iterated integral to polar coordinates and evaluate

$$\int_{-1}^0 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x\ dy\ dx.$$

$$\begin{aligned}
\int_{-1}^0 \int_{-\sqrt{1-x^2}}^{\sqrt{1-x^2}} x\ dy\ dx &= \int_{\pi/2}^{3\pi/2} \int_0^1 r^2\cos(\theta)\ dr d\theta \\
&= \frac{1}{3}\int_{\pi/2}^{3\pi/2} \cos(\theta)\ d\theta \\
&= \frac{1}{3}(\sin(3\pi/2) - \sin(\pi/2)) \\
&= -\frac{2}{3}.
\end{aligned}$$